e-Yantra Robotics Competition 2019-20

Theme Analysis and Implementation

<4078>

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Testing your knowledge (Related to Hardware)

Q1. ___12.5W____ is the Max. Output power delivered by eYFi-Mega board [1 Mark]

<Teams should refer the Hardware Manual provided>

Q2. The Line Follower Array has the default I2C address of 0x20. What changes must be done to change the address to 0x27?
Is R/W bit considered in address formation (Yes/ No)? [2 Marks]

<Teams should refer the Manuals provided. You may use figures / diagrams to support your answer.
Answer format: Text, word-limit: 50 words >

Ans- To change the address to 0x27 from default value 0x20 (0100000_2) move the solder or jumper to set A2, A1, A0 (Fig. 1), according to binary equivalent of 0x27 i.e. 0100111. Comparing the last three bits, move A2, A1 to the '1' position and same for A0 at '1'.

No, R/W bit is not considered in address information. It is separate and used as a control byte.

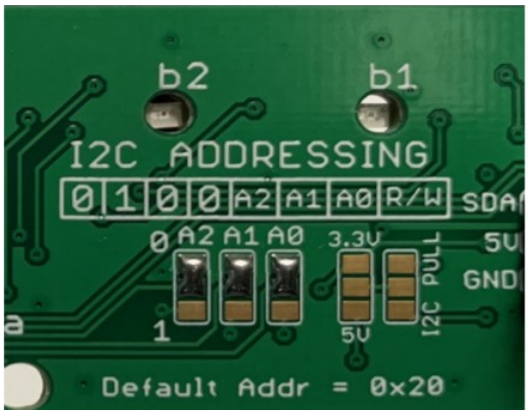


Fig. 1: I2C Addressing

Q3. Universal Asynchronous Receiver and Transmitter (UART) protocol is used for communication between ATmega2560 and ESP32 on eYFi-Mega board.

[1 Mark]

<Teams should refer the Hardware Manual provided>

Q4. 7.0 to 21.0V is the voltage range of External DC Power Supply for eYFi-Mega board.

2.5A is the max. current delivered by 5V when powered through External Supply.

800mA is the max. current delivered by 3.3V when powered through External Supply.

[3 Marks]

<Teams should refer the Manuals provided>

Q5. Where are you planning to place the eY-LFA sensor on the robot. Justify your answer.

[5 Marks]

<Answer format: Text, word-limit: 200 words>

Ans: We intend to place the eY-LFA sensor just at the front. The reason being that, as soon as the bot is out to take victims to hospital, it should be able to follow the line that leads to the hospital.
If we place it anywhere else, the bot will not be able to follow the line when it reaches at the end of inner area.

Q6. How many VL53L0X sensors are you planning to use and where you will place them on the robot. Justify your answer. [6 Marks]

<Answer format: Text, word-limit: 200 words>

Ans: We are planning to use all three of the VL53L0X sensors, the intended placement will have one sensor at a height of 14cm facing front, one at the left side, and another at the front at ground level. Reason being that, wall following only needs one sensor and that is what the sensor at left will be there for. But, for any wall at front, we need a sensor at front, and at ground level so that any wall can be avoided. But in order to differentiate it from ordinary wall we need a sensor at 14cm (<15 cm) as that is the only difference apart from color.

Testing your knowledge (Related to Rulebook)

Q1. Consider the following scenario for all the below cases:
Total Number of Vacancies in all Hospitals (X) = 12
Digits in the maze image = [2, 4, 3, 6, 7, 1, 5, 8]
Total Time taken to complete the task (T) = 200 sec
Assume the cases mentioned below, consider the locations of digits as dummy and the shortest path is communicated to robot using which it tries to traverse all the firezones planned.

Case I : [4 Marks]

a) **All Digits in image = # [2, 4, 3, 6, 7, 1, 5, 8] #**
b) **Combination of digits for sum with their locations = # { 2: (2, 4), 3: (3, 5), 7: (3, 7) } #**
c) **Robot visited all planned firezones correctly**

Case II : [4 Marks]

a) **All Digits in image = # [2, 4, 3, 6, 7, 1, 6, 8] #**
b) **Combination of digits for sum with their locations = # { 7: (3, 7), 4: (4, 8) } #**
c) **Robot only missed to visit the firezone at location (3, 7)**

Case III : [4 Marks]

a) **All Digits in image = # [2, 4, 3, 6, 7, 1, 5, 8] #**
b) **Combination of digits for sum with their locations = # { 6: (4, 4), 7: (3, 7) } #**
c) **Robot visited all planned firezones correctly**

Fill in the values of parameters of the Scoring Formula for each case.
<Teams should refer the Rulebook provided>

Parameters	Case I	Case II	Case III
VFC	12	4	12
VFI	0	7	0
NFP	3	2	2
NVP	12	11	13
NVC	3	1	2
NEC	0	0	0
NVH	1	0	1
B	**1**	**0**	**1**
Total	**1850**	**425**	**1800**

<u>IMPORTANT</u>:

- **The document you submit should be in YOUR OWN WORDS. To avoid any copyright violations, you must NOT copy phrases directly from manuals or web.**
- **The team should NOT mail or upload the document anywhere else except on the portal.**
- **Teams failing to submit the document by the deadline will lose the marks for this task.**
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